

GECON 7201

Applied Econometrics

Assignment 1
Due Thursday October 8, 2026

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Instructions

- Edit this Quarto document for your assignment. All answers must be typed in this document. Answers are to be written in the callout blocks (see below)
- Provide a link to your website in Nexus, so that I can view your assignment.
- Upload both a PDF and Quarto file to the nexus assignment portal.
- The use of AI is strictly prohibited for this assignment.
- Show your work
- Type your mathematical answers using LaTeX
- For questions marked R, include both your code and its output.
- You may use `solve()`, `t()`, `%*%`, `diag()`, `qr()`, `eigen()`, and other base R functions freely unless a question says otherwise.
- Please round numerical answers to two decimal places unless stated otherwise.

```
set.seed(3201)
```

Questions

Q1

Create a personal Quarto website hosted on GitHub pages. For reference, please see the slides for **website** on Nexus. Your website should include the following:

- a. An “Index” or “Home” page. This page should include the following: - Your name - Your photo (optional) - Your credentials - An “About Me” section that provides a brief (one paragraph) introduction to yourself. [10 Marks]

- b. A blog page. This will be the landing site for future assignments [5 Marks]
- c. Create a blog post and upload the finished PDF for this assignment. [3 Marks]
- d. Navigation bar with “Home” and “Blog” buttons. The navigation bar is edited in the “_quarto.yml” file. [2 Marks]

Paste your link here

Q2

Consider

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & -3 \\ 0 & -3 & 6 \end{bmatrix}.$$

- a. State the dimensions of A and identify a_{23} . [2 Marks]

Answer

Matrix A has 3 row and 3 column. The dimension of the matrix A is (3×3) . a_{23} means the element of the matrix A is situated in row 2 and column 3 which vale we can write as

$$a_{23} = -1$$

- b. Is A a square matrix? Is it symmetric? Justify your answer using the definition $A = A^T$, not just by inspection. [2 Marks]

Answer

A square matrix is a matrix which has an equal number of rows and columns. Here we can matrix A has 3 rows and 3 columns (3×3) . So Martix A is square matrix.

A matrix is symmetric matrix if and only if the transpose of the matrix same as the original matrix. Here

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix},$$

The Transpose of matrix A is

$$A^T = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix},$$

We can see $A = A^T$. SO, Matrix A is symmetric.

c. Is A a diagonal matrix? Explain why or why not. [1 Mark]

Answer

A matrix is a diagonal matrix if all the outside element of the main diagonal is zero. Here we can see

$$a_{13} = 1, a_{23} = -1, a_{31} = 1, a_{32} = -1$$

But we can see here A matrix all outside element is nonzero. SO Matrix A is not a diagonal matrix.

d. In R, Enter A and B into R. Use `isSymmetric()` to confirm your answer to part (b) for both matrices. [4 Marks]

Answer

```
A <- matrix(c(2, 0, 1,
              0, 3, -1,
              1, -1, 4),
            nrow = 3, byrow = TRUE)
```

```
B <- matrix(c(1, 2, 0,
              2, 5, -3,
              0, -3, 6),
            nrow = 3, byrow = TRUE)
```

```
isSymmetric(A)
```

```
[1] TRUE
```

```
isSymmetric(B)
```

```
[1] TRUE
```

Both matrix `isSymmetric()` is **True**. So both matrix A and B are symmetric matrix. Answer (b) is justified.

Q3

Using the same A and B from Question 1:

a. Compute $A + B$ and $3A - 2B$ by hand. [4 marks]

Answer

$$A + B = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 8 & -4 \\ 1 & -4 & 10 \end{bmatrix},$$

and Here we calculate $3A - 2B$

$$3A - 2B = \begin{bmatrix} 4 & -4 & 3 \\ -4 & -1 & 3 \\ 3 & 3 & 0 \end{bmatrix}$$

b. Compute AB by hand, showing at least one entry's calculation in full using the summation formula $(AB)_{ij} = \sum_{\ell} a_{i\ell}b_{\ell j}$. [2 marks]

Answer

Using the summation formula

$$(AB)_{ij} = \sum_{\ell} a_{i\ell}b_{\ell j}$$

for one entry calculation we can use this

$$\begin{aligned} (AB)_{11} &= a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} \\ (AB)_{11} &= (2)(1) + (0)(2) + (1)(0) \\ (AB)_{11} &= 2 \end{aligned}$$

$$AB = \begin{bmatrix} 2 & 1 & 3 \\ 6 & 18 & -15 \\ -1 & -15 & 27 \end{bmatrix},$$

c. Compute BA . Is $AB = BA$ here? Was this guaranteed in general? [2 marks]

$$BA = \begin{bmatrix} 2 & 6 & -1 \\ 1 & 18 & -15 \\ 6 & -15 & 27 \end{bmatrix},$$

SO we can see $AB \neq BA$. so it is not guaranteed.

d. In R, verify parts (a)–(c) in R. Also compute $A * B$ (element-wise) and explain in one sentence why this is a different operation from $A \%*\% B$. [4 marks]

Answer

```
A <- matrix(c(2, 0, 1,
              0, 3, -1,
              1, -1, 4),
            nrow = 3, byrow = TRUE)
```

```
B <- matrix(c(1, 2, 0,
              2, 5, -3,
              0, -3, 6),
            nrow = 3, byrow = TRUE)
```

```
# (a) A+B
```

```
A+B
```

```
      [,1] [,2] [,3]
[1,]     3     2     1
[2,]     2     8    -4
[3,]     1    -4    10
```

```
# 3A-2B
```

```
3*A - 2*B
```

```
      [,1] [,2] [,3]
[1,]     4    -4     3
[2,]    -4    -1     3
[3,]     3     3     0
```

```
# (b) AB
```

```
A%*%B
```

```
      [,1] [,2] [,3]
[1,]     2     1     6
[2,]     6    18   -15
[3,]    -1   -15    27
```

```
# (c) BA
```

```
B%*%A
```

```

      [,1] [,2] [,3]
[1,]    2    6   -1
[2,]    1   18  -15
[3,]    6  -15   27

```

```
# AB=BA
```

```
all(A%%B==B%%A)
```

```
[1] FALSE
```

```
# A*B
```

```
A*B
```

```

      [,1] [,2] [,3]
[1,]    2    0    0
[2,]    0   15    3
[3,]    0    3   24

```

In $A * B$ each element of A is multiplied by the corresponding element of B . and In A each element is calculated by using row by column multiplication. SO, $A * B$ and A operation is different.

Q4

Let

$$X = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

a. Write down X^T and state its dimensions. [2 marks]

Answer

$$X^T = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}.$$

Matrix X^T has 2 row and 3 columns. if we take any matrix transpose ,its switch the dimension of the original matrix. So A matrix has the dimension (3×2) and X^T matrix dimension is (2×3) .

- b. Compute $X^T X$ and XX^T by hand. Explain why these two products have different dimensions even though both involve X and X^T . [4 marks]

Answer

Computing $X^T X$,

$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}$$

Above $X^T X$ matrix dimension is (2×2) . Now we compute XX^T ,

$$\mathbf{X} \mathbf{X}^T = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix}$$

Above XX^T matrix dimension is (3×3) . Though both matrix involve X and X^T , they have different dimension because of matrix multiplication depends on dimension of the matrix. Such as $XX^T = (2 \times 3)(3 \times 2) = (2 \times 2)$ and $X^T X = (3 \times 2)(2 \times 3) = (3 \times 3)$

- c. Let $\mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$. Compute $\mathbf{v}^T \mathbf{v}$ and $\mathbf{v} \mathbf{v}^T$ by hand, and state the dimension of each result. [4 marks]

Answer

$$\mathbf{v}^T = [1 \quad -1 \quad 2]$$

Matrix $\mathbf{v}^T$ has 1 row and 3 column (1×3) . Now we compute $\mathbf{v}^T \mathbf{v}$. Using Matrix multiplication

$$\mathbf{v}^T \mathbf{v} = [6]$$

This matrix Has 1 row and 1 column (1×1) . Also it has only 1 value which is 6. that's why it is also called scalar.

Now We will find

$$\mathbf{v} \mathbf{v}^T = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{bmatrix}$$

This matrix Has 3 row and 3 columns. The dimension of the matrix is (3×3) .

- d. In R, verify parts (a)–(c). Then confirm numerically, for two conformable matrices of your choosing, that $(AB)^T = B^T A^T$. [4 marks]

Answer

```
# (a)
X <- matrix(c(1, 4,
              2, 5,
              3, 6),
            nrow = 3, byrow = TRUE)
# verifying Transpose of X
t(X)
```

```
      [,1] [,2] [,3]
[1,]     1     2     3
[2,]     4     5     6
```

```
# small d is the dimension of the transpose of X
X_transpose <- t(X)
#verifying Dimension
dim(X_transpose)
```

```
[1] 2 3
```

```
# (b)
#verifying X_transpose X and X X_transpose
X_transpose %*% X
```

```
      [,1] [,2]
[1,]    14    32
[2,]    32    77
```

```
X %*% X_transpose
```

```
      [,1] [,2] [,3]
[1,]    17    22    27
[2,]    22    29    36
[3,]    27    36    45
```

```
#verifying dimension of X_transpose %*% X
dim(X_transpose %*% X)
```

```
[1] 2 2
```



```
#verifying Dimension of X %*% X_transpose
dim(X %*% X_transpose)
```

```
[1] 3 3
```

```
# (c)
V <- matrix(c(1 ,
              -1,
              2),
            nrow = 3, byrow = TRUE)
```

```
# Calculating Transpose of V
t(V)
```

```
      [,1] [,2] [,3]
[1,]     1  -1    2
```

```
V_transpose <- t(V)
# Verifying V_transpose V
```

```
V_transpose %*% V
```

```
      [,1]
[1,]     6
```

```
#Verifying the Dimension of V_transpose V
dim(V_transpose %*% V)
```

```
[1] 1 1
```

```
#Verifying V V_transpose
```

```
V %*% V_transpose
```

```
      [,1] [,2] [,3]
[1,]     1  -1    2
[2,]    -1    1   -2
[3,]     2  -2    4
```

```
#Verifying the Dimension of V %*% V_transpose
```

```
dim(V %*% V_transpose)
```

```
[1] 3 3
```

```
# verifying (AB)_transpose = B_transpose A_transpose
```

```
A <- matrix(c(3, 5,  
             1, 7),  
            nrow = 2, byrow = TRUE)
```

```
B <- matrix(c(5, 13,  
             2, 7),  
            nrow = 2, byrow = TRUE)
```

```
A %*% B
```

```
      [,1] [,2]  
[1,]    25    74  
[2,]    19    62
```

```
AB <- A %*% B
```

```
# Calculating AB_tranpose
```

```
t(AB)
```

```
      [,1] [,2]  
[1,]    25    19  
[2,]    74    62
```

```
AB_transpose <- t(AB)
```

```
t(A)
```

```
      [,1] [,2]  
[1,]     3     1  
[2,]     5     7
```

```
A_transpose <- t(A)
```

```
t(B)
```

```

      [,1] [,2]
[1,]    5    2
[2,]   13    7

```

```
B_transpose <- t(B)
```

```
#now Calculating B_transpose A_transpose
```

```
B_transpose %*% A_transpose
```

```

      [,1] [,2]
[1,]   25   19
[2,]   74   62

```

```
B_transposeA_transpose <- B_transpose %*% A_transpose
```

```
# verifying (AB)_transpose = B_transpose A_transpose
```

```
all(AB_transpose == B_transposeA_transpose)
```

```
[1] TRUE
```

Q5

Consider

$$A = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}.$$

- a. In R, compute A^{-1} using `solve()`. [2 marks]

Answer

```

A <- matrix(c(4, 2,
              3, 3),
            nrow = 2, byrow = TRUE)
solve(A)

```

```

      [,1]      [,2]
[1,]  0.5 -0.3333333
[2,] -0.5  0.6666667

```

- b. Construct a 3×3 matrix that is singular (for example, make one column a linear combination of the others) and show what happens when you call `solve()` on it. [2 marks]

ANswer

```
E <- matrix(c(7, 5, 12,
              2, 3, 5,
              1, 8, 9),
            nrow = 3, byrow = TRUE)
det(E)
```

```
[1] 0
```

```
tryCatch(
  solve(E),
  error = function(e) {
    cat("Error:", e$message)
  }
)
```

Error: Lapack routine dgesv: system is exactly singular: U[3,3] = 0

This matrix is singular that's why its inverse is not possible. (Above r chunk i took help from ChatGPT to stick with error message and run the quarto document otherwise I couldn't run the file)

Q6

- a. Define, in your own words, what it means for a set of vectors to be linearly independent. [1 mark]

ANswer

A set of the vectors is to be linearly independent when none of the vector can be written as the combination of others.

- b. Consider

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}.$$

Show that $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are linearly dependent by finding constants c_1, c_2, c_3 , not all zero, such that $c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3 = \mathbf{0}$. [3 marks]

Answer

Given equation $c_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3 = \mathbf{0}$. substituting the value of the vector into the given equation we get,

$$\begin{aligned}c_1 + 2c_3 &= 0 \\2c_1 + c_2 + 5c_3 &= 0 \\c_1 + c_2 + 3c_3 &= 0\end{aligned}$$

From the first equation $c_1 + 2c_3 = 0$, we get $c_1 = -2c_3$, Putting this value into equation $2c_1 + c_2 + 5c_3 = 0$, we get $c_2 = -c_3$. As all constraint are not zero, We assume $c_3 = -2$ now we get all constraint value $c_1 = 4, c_2 = -2, c_3 = -2$ now putting all constraints value into given equation we get $2\mathbf{x}_1 + \mathbf{x}_2 - \mathbf{x}_3 = \mathbf{0}$. So we can see all the co-efficient are not zero thus, they are linearly dependent.

- c. Let $M = [\mathbf{x}_1 \ \mathbf{x}_2 \ \mathbf{x}_3]$. What is $\text{rank}(M)$? Explain your reasoning using your answer to part (b). [2 marks]

Answer

From (b), we found that three vectors are linearly dependent. The columns of M are linearly dependent so that M can not be full rank. M has 3 column, so its rank will be less than 3. From (b) equation $2X_1 + X_2 - X_3 = 0$, $X_3 = 2X_1 + X_2$. In this equation we X_3 is dependent on $2X_1 + X_2$ and X_1 and X_2 are independent. SO, The $\text{Rank}(M) = 3 - 1 = 2$

- d. In R, build M and confirm its rank using `qr(M)$rank`. Then modify one entry of $\mathbf{x}_3$ so that the three vectors become linearly independent, and confirm the new rank. [3 marks]

Answer

```
X1 <- c(1, 2, 1)
X2 <- c(0, 1, 1)
X3 <- c(2, 5, 3)
M <- cbind(X1, X2, X3)
M
```

```
      X1 X2 X3
[1,]  1  0  2
```

```
[2,] 2 1 5
[3,] 1 1 3
```

```
#verifying rank of M
qr(M)$rank
```

```
[1] 2
```

Modifying the one entry of X_3

$$\mathbf{x}_3 = \begin{bmatrix} 3 \\ 5 \\ 3 \end{bmatrix}.$$

```
X1 <- c(1, 2, 1)
X2 <- c(0, 1, 1)
X3n <- c(3, 5, 3)
Mn <- cbind(X1, X2, X3n)
Mn
```

```
      X1 X2 X3n
[1,]  1  0   3
[2,]  2  1   5
[3,]  1  1   3
```

```
#verifying rank of Mn
qr(Mn)$rank
```

```
[1] 3
```

The Mn is linearly independent and this is a full rank 3.

Q7.

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

- a. Write out the quadratic form $\mathbf{x}^\top A \mathbf{x}$ in terms of x_1 and x_2 . [2 marks]

Answer

The quadratic form of $X^T Ax$ is $X^T Ax = 2(x_1^2 + x_2^2) + 2x_1x_2$

- b. Is B invertible? Justify your answer using the relationship between definiteness, rank, and invertibility discussed in class. [3 marks]

Answer

In B both columns are identical. So The columns are linearly dependent. Therefore its $\text{rank}(B) = 2 - 1 = 1$. Since B doesn't have full rank its $\det(B) = 0$ so this is a singular matrix. Here $X^T Bx = (x_1 + x_2)^2 \geq 0$ This is positive semi-definite. In conclusion, B is linearly dependent $\rightarrow B$ doesn't have full column rank $\rightarrow B$ is positive semi-definite $\rightarrow B$ is not invertible.

- c. In R, Confirm parts (a) and (b) by computing `eigen(A)$values` and `eigen(B)$values`. State the rule connecting the sign of a symmetric matrix's eigenvalues to whether it is positive definite, positive semidefinite, or neither. [4 marks]

Answer

```
A <- matrix(c(2, 1,
              1, 2),
            nrow = 2, byrow = TRUE)
B <- matrix(c(1, 1,
              1, 1),
            nrow = 2, byrow = TRUE)
#Finding eigenvalues
eigen(A)$values
```

```
[1] 3 1
```

```
eigen(B)$values
```

```
[1] 2 0
```

For a symmetric matrix if all eigenvalues are positive, the matrix is positive definite. If all eigenvalues are non-negative like at least one of the eigenvalues is zero then the matrix is positive semi-definite. If eigenvalues are at least one negative and at least one positive, then neither of the positive and semi-positive definite. So we can see A Matrix has both positive eigenvalues. This is positive definite. B matrix has at least one eigenvalue is zero. This is positive semi-definite.

Q8

Let

$$A = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}.$$

- a. Show by direct calculation that $A^2 = A$, i.e., that A is idempotent. [2 marks]

Answer

$$A^2 = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$$

$A^2 = A$ So This is idempotent matrix.

- b. Using only the idempotent property $A^2 = A$ (not this specific matrix), show that $I - A$ is also idempotent. [2 marks]

Answer

A is Idempotent, So $A^2 = A$ Let $I - A$ Idempotent

$$\begin{aligned} (I - A)^2 &= I - 2A + A^2 \\ (I - A)^2 &= I - A \end{aligned}$$

SO, $I - A$ is Idempotent.

- c. In R, confirm that A is idempotent using matrix multiplication in R. [1 mark]

Answer

```
A <- matrix(c(0.5, 0.5,
              0.5, 0.5),
            nrow = 2, byrow = TRUE)
```

A

```
      [,1] [,2]
[1,]  0.5  0.5
[2,]  0.5  0.5
```

```
A%*%A
```

```
      [,1] [,2]
```



```
[1,] 0.5 0.5
[2,] 0.5 0.5
```

```
#checking A%%A = A
all(A%%A==A)
```

```
[1] TRUE
```

Q9

Let $\mathbf{a} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and

$$S = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Note that S is symmetric.

- a. Write down $\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}}$ for a general 3×1 vector $\mathbf{x}$. (No calculation needed, state the rule and apply it to this $\mathbf{a}$.) [2 marks]

Answer

$$\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}} = \mathbf{a} \text{ Given } \mathbf{a} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} \text{ So, } \frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}} = \mathbf{a} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$$

- b. Derive $\frac{\partial(\mathbf{x}^\top S \mathbf{x})}{\partial \mathbf{x}}$ using the general rule for $\frac{\partial(\mathbf{x}^\top A \mathbf{x})}{\partial \mathbf{x}}$, and simplify using the fact that S is symmetric. [2 marks]

Answer

Given $\frac{\partial(\mathbf{x}^\top A \mathbf{x})}{\partial \mathbf{x}} = (A + A^\top) \mathbf{x}$ Here $A=S$, $\frac{\partial(\mathbf{x}^\top S \mathbf{x})}{\partial \mathbf{x}} = (S + S^\top) \mathbf{x}$

If S is symmetric, $S^\top = S$

$$(S + S^\top) \mathbf{x} = (S + S) \mathbf{x} = 2S \mathbf{x}$$

So we can write the derived form $\frac{\partial(\mathbf{x}^\top S \mathbf{x})}{\partial \mathbf{x}} = 2S \mathbf{x}$

c. Evaluate your answer to part (b) at $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$. [1 mark]

Answer

$$\mathbf{S}\mathbf{x} = \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix} \quad 2\mathbf{S}\mathbf{x} = 2 \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix} \quad 2\mathbf{S}\mathbf{x} = \begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix}$$

d. In R, using the `numDeriv` package (`grad()`), numerically confirm your answers to parts (a) and (c). [2 marks]

answer

```
library(numDeriv)

# (a)
a <- c(3, -2, 1)
x <- c(1, 0, -1)
f_lin <- function(x)
  sum(a * x)
cbind(grad(f_lin, x), a)
```

```
      a
[1,]  3  3
[2,] -2 -2
[3,]  1  1
```

```
# (b)
s <- matrix(c(2, 1, 0,
              1, 3, 1,
              0, 1, 2),
            nrow = 3, byrow = TRUE)
f_quad <- function(x) as.numeric(t(x) %*% s %*% x)
cbind(grad(f_quad, x), (s+t(s)) %*% x)
```

```
      [,1] [,2]
[1,]  4.000000e+00    4
[2,]  1.566452e-15    0
[3,] -4.000000e+00   -4
```

Q10

Let $\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$ be a random vector with

$$\mu = E[\mathbf{X}] = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \quad \Sigma = \text{Var}(\mathbf{X}) = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}.$$

Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$, and define $\mathbf{Y} = A\mathbf{X} + \mathbf{b}$.

a. State $E[\mathbf{Y}]$ and $\text{Var}(\mathbf{Y})$, citing the properties used. [3 marks]

Answer

Given $\mathbf{Y} = A\mathbf{X} + \mathbf{b}$.

$E[A\mathbf{X} + \mathbf{b}] = AE[\mathbf{X} + \mathbf{b}]$ using the above property, we can write $E[\mathbf{Y}] = E[A\mathbf{X} + \mathbf{b}]$ so , we can write, $E[\mathbf{Y}] = AE[\mathbf{X}] + E[\mathbf{b}]$ So, $E[\mathbf{Y}] = A\mu + \mathbf{b}$

Variance

$\text{Var}(A\mathbf{X} + \mathbf{b}) = A \text{Var}(\mathbf{X}) A^\top$ using the property we can write $\text{Var}(A\mathbf{X} + \mathbf{b}) = A \Sigma A^\top$

b. Compute $E[\mathbf{Y}]$ by hand. [2 marks]

Answer

$$E[\mathbf{Y}] = \begin{bmatrix} 7 \\ -1 \end{bmatrix}$$

c. Compute $\text{Var}(\mathbf{Y}) = A \Sigma A^\top$ by hand, showing your intermediate matrix products. [3 marks]

answer

Computing $A \Sigma$

$$A \Sigma = \begin{bmatrix} 5 & 3 & -1 \\ 1 & 3 & -4 \end{bmatrix}$$

Now we compute $A^\top = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & -1 \end{bmatrix}$

So, $A \Sigma A^\top = \begin{bmatrix} 8 & 4 \\ 4 & 7 \end{bmatrix}$

d. Let $\mathbf{a} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$. Compute $\text{Var}(\mathbf{a}^\top \mathbf{X})$ using the appropriate property, without recomputing $\text{Var}(\mathbf{X})$ from scratch. [3 marks]

Answer

$\text{Var}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \Sigma \mathbf{a}$ Calculating $\Sigma \mathbf{a} = \begin{bmatrix} 4(1) + 1(-1) + 0(2) \\ 1(1) + 2(-1) + (-1)(2) \\ 0(1) + (-1)(-1) + 3(2) \end{bmatrix}$

so $\Sigma \mathbf{a} = \begin{bmatrix} 3 \\ -3 \\ 7 \end{bmatrix}$

now calculating $\mathbf{a}^\top = [1 \quad -1 \quad 2]$

Now we get $\text{Var}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \Sigma \mathbf{a} = 1(3) + (-1)(-3) + 2(7) = 20$

e. In R, using `MASS::mvrnorm()`, draw 100,000 observations from a distribution with mean μ and variance Σ (you may treat $\mathbf{X}$ as multivariate normal for this simulation). Confirm your answers to (b), (c), and (d) using `colMeans()` and `cov()`. [3 marks]

Answer

```

library(MASS)
mu <- c(2, 0, -1)
sigma <- matrix(c(4, 1, 0,
                  1, 2, -1,
                  0, -1, 3),
                nrow = 3, byrow = TRUE)
A <- matrix(c(1, 1, 0,
              0, 1, -1),
            nrow = 2, byrow = TRUE)
b <- c(5, -2)
# generating 100,000 observations
x <- mvrnorm(n=100000, mu = mu, Sigma = sigma)
# Y=AX+b, one row per draw
Y <- x%*% t(A) +matrix(b, nrow(x), 2, byrow = TRUE)

# (b) checking E[Y]
colMeans(Y)

```

```
[1] 6.9938519 -0.9968664
```

```

# (c) checking Var(Y)
cov(Y)

```

```

      [,1]      [,2]
[1,] 8.016676 3.991698
[2,] 3.991698 7.009904

```

```

# (d) checking var(a_transpose x)

a <- c(1, -1, 2)
Z <- x %*% a
var(Z)

```

```

      [,1]
[1,] 20.0895

```

Q11

- a. Suppose $\mathbf{X} \sim N_3(\mu, \Sigma)$ using the μ and Σ from Question 10. What is the marginal distribution of X_2 alone? Give its distribution and parameters. [2 marks]

Answer

Here $mean = \mu$ and $variance = \sum X_2$ follows this normal distribution $\mathbf{X} \sim N_3(\mu, \Sigma)$ For $X_2 = \mu = 0$ and $\sum_{22} = 2$ So, we can write $\mathbf{X} \sim N_3(0, 2)$

- c. Using property 4 of the multivariate normal (zero covariance implies independence), determine whether X_1 and X_3 are independent. Justify your answer using an entry of Σ . [1 mark]

Answer

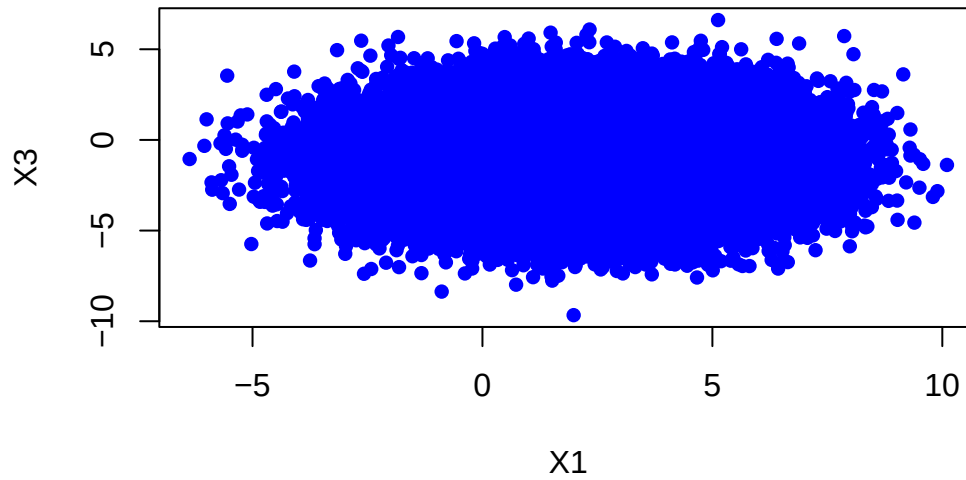
According to the property four :Zero covariance implies independence(If $Cov(\mathbf{X}_i, \mathbf{X}_j) = 0$ then $\mathbf{X}_i$ and $\mathbf{X}_j$ are independent.) Using Covariance matrix we got $Cov(\mathbf{X}_1, \mathbf{X}_3) = \sum_{13} = 0$
So we can according to the property four $\mathbf{X}_1, \mathbf{X}_3$ are independent.

- d. In R, using your simulated draws from Question 10(e), produce a scatterplot of X_1 against X_3 and compute `cor(X1, X3)`. Comment on whether the plot is consistent with your answer to part (c). [3 marks]

Answer

```
plot(x[,1], x[,3],  
     xlab = "X1",  
     ylab = "X3",  
     main = "Scatterplot of X1 against X3",  
     pch = 16 ,  
     col = "Blue")
```

Scatterplot of X1 against X3



```
cor(x[,1], x[,3])
```

```
[1] 0.001875436
```